

Team 02 Database Schema

Good Driver Incentive Program

Database schema based on our ERD, project requirements, and user stories.

USER_ACCOUNT

Login/profile information shared by drivers, sponsor users, and admins.

Key	Column	Type	Notes
PK	user_id	BIGINT	Internal user ID.
UQ	username	VARCHAR(100)	Unique login.
	password_hash	VARCHAR(255)	Hashed password.
	role	VARCHAR(20)	DRIVER, SPONSOR_USER, ADMIN.
	first_name	VARCHAR(100)	First name.
	last_name	VARCHAR(100)	Last name.
UQ	email	VARCHAR(255)	Email.
	phone	VARCHAR(30) NULL	Phone.
	is_active	BOOLEAN	Account status.
	created_at	TIMESTAMP	Created date.

SPONSOR

Sponsor company information.

Key	Column	Type	Notes
PK	sponsor_id	BIGINT	Sponsor ID.
	sponsor_name	VARCHAR(200)	Company name.
	point_dollar_rate	DECIMAL(10,4)	Default 0.01 per point.
	status	VARCHAR(20)	ACTIVE / INACTIVE.
	contact_email	VARCHAR(255) NULL	Contact email.
	created_at	TIMESTAMP	Created date.

DRIVER

Driver record. sponsor_id enforces one sponsor for each participating driver.

Key	Column	Type	Notes
PK/FK	driver_id	BIGINT	References USER_ACCOUNT(user_id).
FK	sponsor_id	BIGINT NOT NULL	References SPONSOR(sponsor_id).
	current_points	BIGINT	Current balance.
	participation_status	VARCHAR(20)	APPLIED, ACTIVE, DROPPED, etc.
	joined_at	TIMESTAMP NULL	Date joined.

SPONSOR_USER

Sponsor employee/login. A sponsor can have multiple sponsor users.

Key	Column	Type	Notes
PK/FK	sponsor_user_id	BIGINT	References USER_ACCOUNT(user_id).
FK	sponsor_id	BIGINT NOT NULL	References SPONSOR(sponsor_id).
	job_title	VARCHAR(100) NULL	Optional job title.

DRIVER_APPLICATION

Stores driver applications and sponsor/admin decisions.

Key	Column	Type	Notes
PK	application_id	BIGINT	Application ID.
FK	driver_id	BIGINT	References DRIVER.
FK	sponsor_id	BIGINT	References SPONSOR.
FK	decided_by_user_id	BIGINT NULL	User who made decision.
	submitted_at	TIMESTAMP	Submitted date.
	status	VARCHAR(20)	PENDING / ACCEPTED / REJECTED.
	decision_at	TIMESTAMP NULL	Decision date.
	reason	VARCHAR(500) NULL	Reason for decision.

PRODUCT

Product information pulled from the public product API.

Key	Column	Type	Notes
PK	product_id	BIGINT	Product ID.
	api_source	VARCHAR(50)	API/provider.
	external_product_id	VARCHAR(200)	Provider product ID.
	product_name	VARCHAR(255)	Name.
	description	TEXT NULL	Description.
	dollar_price	DECIMAL(12,2) NULL	Latest API price.
	availability	VARCHAR(50)	Availability.
	image_url	VARCHAR(1000) NULL	Image.
	category	VARCHAR(100) NULL	Category.
	content_rating	VARCHAR(10)	G / PG.
	last_refreshed_at	TIMESTAMP NULL	Last API refresh.

CATALOG_ITEM

Connects products to a sponsor's catalog.

Key	Column	Type	Notes
PK	catalog_item_id	BIGINT	Catalog item ID.
FK	sponsor_id	BIGINT	References SPONSOR.
FK	product_id	BIGINT	References PRODUCT.
	point_price	BIGINT	Price in points.
	is_active	BOOLEAN	Currently offered.
	added_at	TIMESTAMP	Added date.
UQ	sponsor_id + product_id	COMPOSITE	No duplicate product in same sponsor catalog.

SHOPPING_CART / CART_ITEM

Stores a driver's shopping cart.

Key	Column	Type	Notes
PK	cart_id	BIGINT	Cart ID.
FK	driver_id	BIGINT	References DRIVER.
	status	VARCHAR(20)	ACTIVE / CHECKED_OUT / ABANDONED.
	created_at / updated_at	TIMESTAMP	Cart dates.
PK/FK	cart_id + catalog_item_id	COMPOSITE	CART_ITEM key.
	quantity	INT	Quantity.
	unit_points_at_add	BIGINT	Point price when added.
	checked_at	TIMESTAMP NULL	Last price/availability check.

CUSTOMER_ORDER / ORDER_ITEM

Stores purchases and historical prices for order status and reports.

Key	Column	Type	Notes
PK	order_id	BIGINT	Order ID.
FK	driver_id	BIGINT	Driver paying with points.
FK	sponsor_id	BIGINT	Sponsor.
FK	placed_by_user_id	BIGINT	User placing order.
	status	VARCHAR(30)	Order status.

Key	Column	Type	Notes
	total_points	BIGINT	Points charged.
	total_dollar_amount	DECIMAL(12,2)	Dollar total.
	placed_at / cancelled_at	TIMESTAMP	Order dates.
PK	order_item_id	BIGINT	Order item ID.
FK	catalog_item_id	BIGINT	Purchased item.
	quantity	INT	Quantity.
	unit_point_price	BIGINT	Point price at purchase.
	unit_dollar_price	DECIMAL(12,2)	Dollar price at purchase.

POINT_TRANSACTION

Stores every point change. A reason is recorded for point additions/deductions.

Key	Column	Type	Notes
PK	point_transaction_id	BIGINT	Transaction ID.
FK	driver_id	BIGINT	References DRIVER.
FK	sponsor_id	BIGINT	References SPONSOR.
FK	performed_by_user_id	BIGINT NULL	User making change.
FK	order_id	BIGINT NULL	Related purchase/refund.
FK	recurring_schedule_id	BIGINT NULL	Related recurring award.
	points_delta	BIGINT	Positive adds; negative deducts.
	balance_after	BIGINT	Balance after change.
	reason	VARCHAR(500)	Reason.
	created_at	TIMESTAMP	Change date.

RECURRING_POINT_SCHEDULE

Supports recurring point awards for good performance.

Key	Column	Type	Notes
PK	schedule_id	BIGINT	Schedule ID.
FK	driver_id	BIGINT	Target driver.
FK	sponsor_id	BIGINT	Sponsor.
FK	created_by_user_id	BIGINT	User creating schedule.
	points_amount	BIGINT	Points each run.
	reason	VARCHAR(500)	Reason.
	frequency	VARCHAR(30)	Frequency.
	next_run_at	TIMESTAMP	Next run.
	is_active	BOOLEAN	Active status.

ALERT_PREFERENCE / NOTIFICATION

Stores driver alert settings and notifications.

Key	Column	Type	Notes
PK/FK	driver_id	BIGINT	Alert preference for DRIVER.
	point_change_enabled	BOOLEAN	Can enable/disable.
	order_enabled	BOOLEAN	Can enable/disable.
PK	notification_id	BIGINT	Notification ID.
FK	user_id	BIGINT	Recipient.
	notification_type	VARCHAR(40)	DROPPED, POINT_CHANGE, ORDER, etc.
	message	TEXT	Message.
	created_at / read_at	TIMESTAMP	Notification dates.

AUDIT_EVENT

Stores required audit information for applications, point changes, password changes, and login attempts.

Key	Column	Type	Notes
PK	audit_event_id	BIGINT	Audit ID.
FK	actor_user_id	BIGINT NULL	User causing event.
FK	sponsor_id	BIGINT NULL	Sponsor involved.
FK	driver_id	BIGINT NULL	Driver involved.

Key	Column	Type	Notes
	category	VARCHAR(40)	APPLICATION / POINT_CHANGE / PASSWORD / LOGIN.
	subject_username	VARCHAR(100) NULL	Username for login events.
	entity_type / entity_id	VARCHAR/BIGINT NULL	Affected record.
	success	BOOLEAN NULL	Success/failure.
	reason_or_details	TEXT NULL	Reason/details.
	created_at	TIMESTAMP	Event date.

ABOUT_RELEASE

Database-backed information for the About page.

Key	Column	Type	Notes
PK	release_id	BIGINT	Release ID.
	team_number	VARCHAR(20)	Team 02.
	version_number	VARCHAR(30)	Sprint/version.
	release_date	DATE	Release date.
	product_name	VARCHAR(200)	Product name.
	product_description	TEXT	Product description.

Relationships

- SPONSOR 1:N DRIVER - each participating driver has one sponsor; one sponsor can have many drivers.
- SPONSOR 1:N SPONSOR_USER - one sponsor can have multiple sponsor users.
- DRIVER / SPONSOR 1:N DRIVER_APPLICATION - keeps application history.
- SPONSOR N:M PRODUCT through CATALOG_ITEM - each sponsor has its own catalog.
- DRIVER 1:N CUSTOMER_ORDER - a driver can have multiple orders.
- CUSTOMER_ORDER 1:N ORDER_ITEM - an order contains one or more items.
- DRIVER 1:N POINT_TRANSACTION - keeps the driver's full point history.
- USER_ACCOUNT 1:N NOTIFICATION - a user can receive multiple notifications.
- USER_ACCOUNT 1:N AUDIT_EVENT - user actions can create audit records.